The Meaning of Interest Rates

Reference: The Economics of Money, Banking, and Financial Markets by Mishkin

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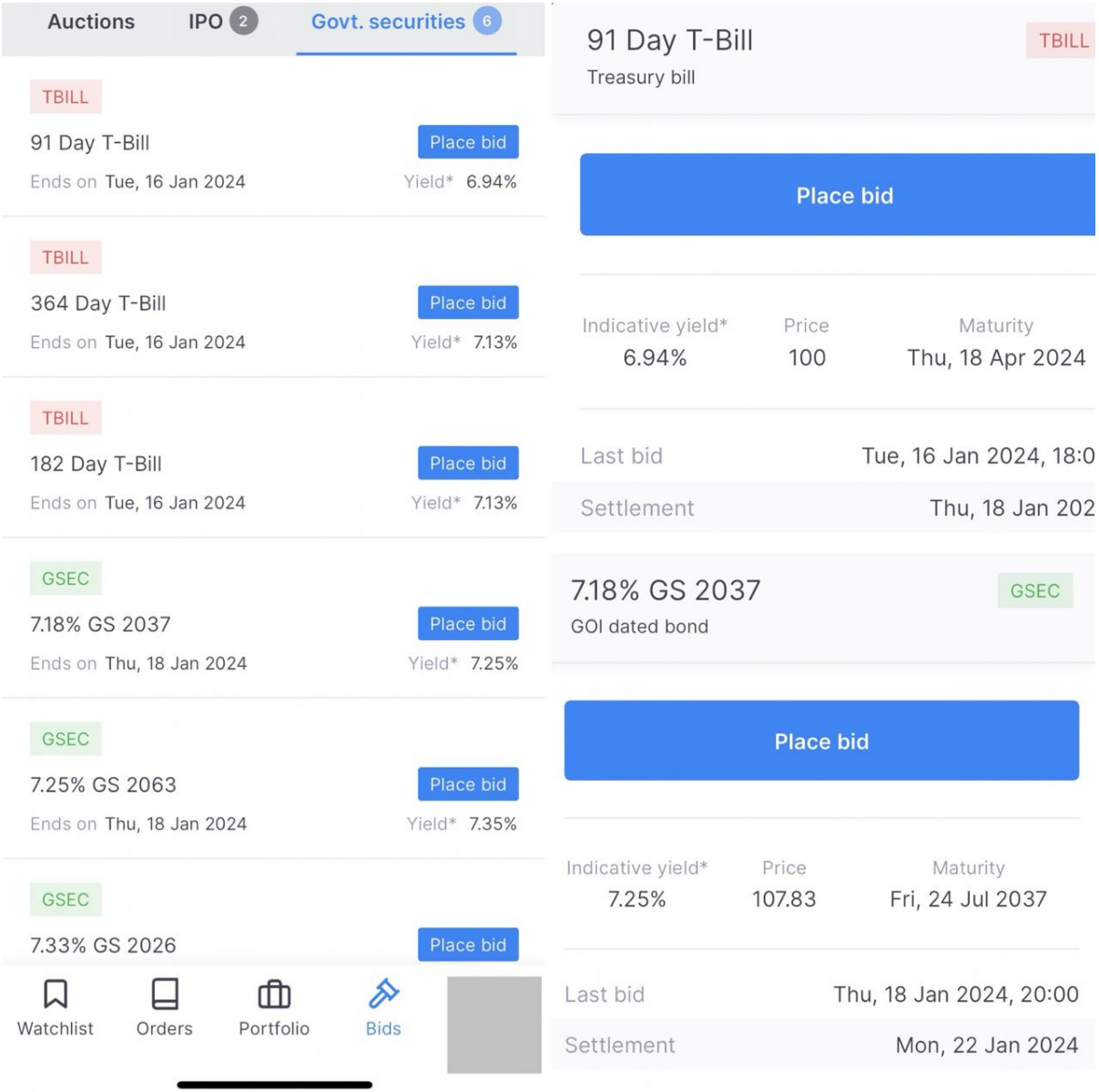
Coupon Bond

- A bond is an instrument of debt, used a lot by governments.
- A coupon bond is identified by four pieces of information:
  - Face value.
  - Agencies that issue this bond.
  - Maturity date.
  - The coupon rate.
- You could say that it hast he flavour of a loan.

Coupon Bond on Kite



Coupon Bond

- The present value of a \$1,000-face-value bond with ten years to maturity and yearly coupon payments of \$100 (a 10% coupon rate) can be calculated as follows:

$$P = \frac{\$100}{1+i} + \frac{\$100}{(1+i)^2} + \frac{\$100}{(1+i)^3} + \ldots + \frac{\$100}{(1+i)^{10}} + \frac{\$1000}{(1+i)^{10}}$$

- The coupon payment, the face value, the years to maturity, and the price of the bond are known quantities, and only the yield to maturity is not.
- Hence we can solve this equation for the yield to maturity i.

Coupon Bond

Table 1 Yields to Maturity on a 10%-Coupon-Rate Bond Maturing in Ten Years (Face Value = \$1,000).

Price of Bond (\$)	Yield to Maturity (%)
1200	7.13
1100	8.48
1000	10.00
900	11.75
800	13.81

Coupon Bond

- When the coupon bond is priced at its face value, the yield to maturity equals the coupon rate.
- The price of a coupon bond and the yield to maturity are negatively related.
- The yield to maturity is greater than the coupon rate when the bond price is below its face value.

Coupon Bond

- Consol or perpetuity: a bond with no maturity date that does not repay principal but pays fixed coupon payments forever.
  - $P_C = \frac{C}{i_C}$.
  - $P_C =$ price of the consol.
  - $C =$ yearly interest payment.
  - $i_C =$ yield to maturity of the consol.
  - One can rewrite the equation as: $i_C = \frac{C}{P_C}$.
- For coupon bonds, this equation gives the current yield, an easy to calculate approximation to the yield to maturity

Coupon Bond

- One nice feature of consols is that you can immediately see that as i goes up, the price of the bond falls.
- For example, if a consol pays \$100 per year forever and the interest rate is 10%,
- Price will be $\$1,000 = \frac{\$100}{0.10}$.

Discount Bond

- For any one year discount bond

$$i = \frac{F - P}{P}$$

- F = Face value of the discount bond.
- P = Current price of the discount bond.
- The yield to maturity equals the increase in price over the year divided by the initial price.
- As with a coupon bond, the yield to maturity is negatively related to the current bond price.

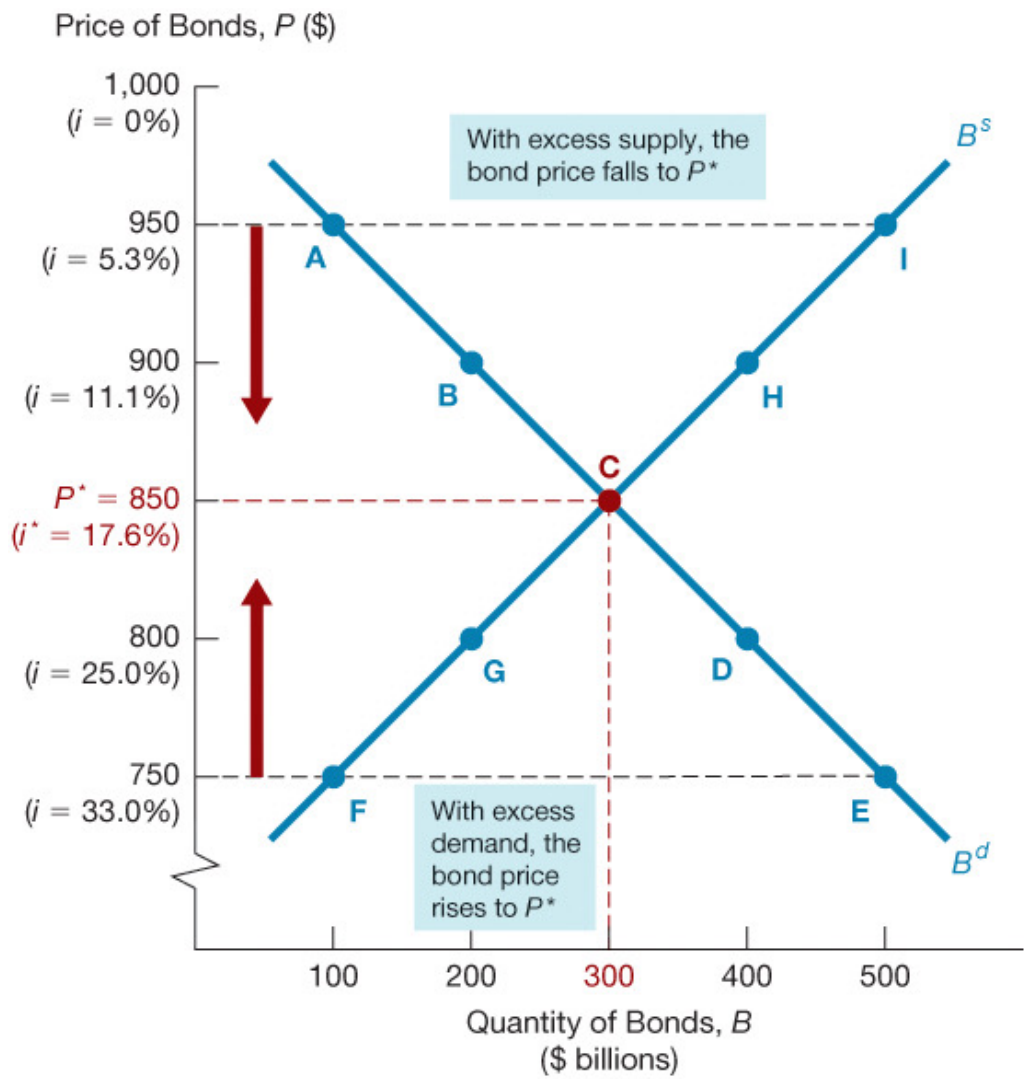
Discount Bond

- Let us consider a discount bond, which pays off a face value of \$1,000 in one year's time.
- If the current purchase price of this bill is \$900.

$$\$900 = \frac{\$1000}{1+i}$$

and solving for i yields $i = 11.1\%$.

Figure 1 Supply and Demand for Bonds



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The Distinction Between Interest Rates and Returns

- Many people think that the interest rate on a bond tells them all they need to know about how well off they are as a result of owning it.
- How well a person does by holding a bond or any other security over a particular time period is accurately measured by the return, or, in more precise terminology, the *rate of return*.

The Distinction Between Interest Rates and Returns

- Let us see what the return would look like for a \$1,000-face-value coupon bond with a coupon rate of 10% that is bought for \$1,000, held for one year, and then sold for \$1,200.
- The payments to the owner are the yearly coupon payments of \$100, and the change in its value is
- \$1,200 - \$1,000 =\$200.
- Adding these together and expressing them as a fraction of the purchase price of \$1,000 gives us the one-year holding-period return for this bond.

The Distinction Between Interest Rates and Returns

- The one-year holding-period return for this bond.

$$\frac{\$100 + \$200}{\$1000} = \frac{\$300}{\$1000} = 0.30 = 30\%.$$

- The return that we have calculated equals 30%,
- The yield to maturity was only 10 percent.
- This demonstrates that the return on a bond will not necessarily equal the interest rate on that bond.

The Distinction Between Interest Rates and Returns

- Rate of return
- The payments to the owner plus the change in value expressed as a fraction of the purchase price

$$\text{RET} = \frac{C}{P_t} + \frac{P_{t+1} - P_t}{P_t}.$$

- RET = return from the holding the bond from time t to time $t+1$.
- P_t = price of bond at time t.
- P_{t+1} = price of bond at time $t+1$.
- C = coupon payment
- $\frac{C}{P_t}$ = current yield = i_C.
- $\frac{p_{t+1}-P_t}{P_t}$ = rate of capital gain = g.

The Distinction Between Interest Rates and Returns

- The more distant a bond's maturity, the greater the size of the percentage price change associated with an interest-rate change.
- The more distant a bond's maturity, the lower the rate of return that occurs as a result of an increase in the interest rate.
- Even if a bond has a substantial initial interest rate, its return can be negative if interest rates rise.

The Distinction Between Real and Nominal Interest Rates

- *Nominal interest rate* makes no allowance for inflation.
- *Real interest rate* is adjusted for changes in price level so it more accurately reflects the cost of borrowing.
  - *Ex ante real interest rate* is adjusted for expected changes in the price level
  - *Ex post real interest rate* is adjusted for actual changes in the price level

Fisher Equation

- Let us first consider a situation in which you have made a one-year simple loan with a 5% interest rate ($i = 5\%$) and you expect the price level to rise by 3% over the course of the year ($\pi^e = 3\%$).
- As a result of making the loan, at the end of the year you will have 2% more in real terms,

Fisher Equation

- Now if the interest rate rises to 8%, but you expect the inflation rate to be 10% over the course of the year?
- Although you will have 8% more dollars at the end of the year, you will be paying 10% more for goods;
- The result is that you will be able to buy 2% fewer goods at the end of the year and you are 2% worse off in real terms.
- When the real interest rate is low, there are greater incentives to borrow and fewer incentives to lend.

Fisher Equation

- A similar distinction can be made between nominal returns and real returns.
- Nominal returns, which do not allow for inflation, are what we have been referring to as simply "returns".
- When inflation is subtracted from a nominal return, we have the real return, which indicates the amount of extra goods and services that can be purchased as a result of holding the security.
- The distinction between real and nominal interest rates is important because the real interest rate, which reflects the real cost of borrowing, is likely to be a better indicator of the incentives to borrow and lend.